Innovative Actuators and Tools for Micro-Nano Mechatronics

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Abstract:
Applications of actuators have increased in various fields. In industry, precise and high speed positioning is one of the most important technologies. In peripheral machines for computers like disc memories, small and thin actuators are necessary to satisfy the demand of reduction of thickness and weight of the products. In some cases conventional actuators seem to be difficult to satisfy these new and advanced demands. Therefore, seeking for new actuators has been activated recently. Especially, in the field of MEMS, where the dominant physical rules are different from those of the usual size, new small actuators utilizing electrostatic force, heat deformation, piezoelectric materials, shape memory, and so on have been proposed and developed. Among the new actuators, impact drive mechanism and surface acoustic wave motor are introduced as examples of piezo actuators. And as examples of actuators and manipulations using electrostatic force, powerful electrostatic motor, thin film electrostatic motor, electrostatic transportation devices of particles and droplets, and electrostatic suspension are described.

1. INTRODUCTION

Many new actuators and drive methods have been developed for motion control needed in automation in industry and in positioning in various kinds of machines and instruments. In our laboratory, a number of unique actuators and drive technologies were invented and realized. Among them, first, the following two actuators using piezoelectric materials, impact drive mechanism and surface acoustic wave motor, are introduced with applications. Impact drive method can move an object with a step of several nm to micrometers for a long distance by using rapid deformation of a piezoelectric element. The surface acoustic wave motor is a very thin ultrasonic motor with promising properties like large thrust, high velocity, and quick response. Then, as applications of electrostatic force, powerful electrostatic motor and new technologies of electrostatic transportation of particles, powder and droplets, sheets and films, and thin plates, are presented respectively with their principles and devices. And combining the technologies of electrostatic suspension without mechanical contact and electrostatic drives, super clean transportation devices for 300 mm and 400mm silicon wafers and thin plates of glass for flat displays are introduced.

2. PIEZO ACTUATORS

In order to get precise positioning with a resolution of from several nanometers to sub microns, piezoelectric elements are the most useful actuator. For instance, the probe of STM, scanning tunneling microscope, should be positioned with the resolution of nm even angstrom. Now only piezoelectric elements can satisfy the requirements commercially. As for the frequency response, a piezo-electric element itself can deform much faster than usual electric linear motors like a voice coil motor. So, the piezoelectric elements seem to be the most convenient actuators when we need to position a rather small object with high accuracy. On the other hand, the maximum deformation of a piezoelectric element itself is limited to very small, like 10 micrometers for a 10 mm long piezoelectric element. Since the displacement of a piezo itself is limited, several methods have been developed to realize longer or boundless movement by combining some mechanism with piezoelectric elements.
The most popular mechanisms are “inchworm” and ultrasonic motors. Since they are well known and have been commonly used, hereafter, two other unique piezoelectric linear actuators are introduced. One is impact drive mechanism, which can be comparing with the inchworm. The other is surface acoustic wave motor, which provides us the thinnest ultrasonic motor.

2.1 Impact Drive Mechanism[1]

To compose an inchworm, at least two active clamps are necessary in addition to a main piezoelectric element. By holding and releasing the object with the clamps in accordance with extension and shrink of the main piezoelectric element, the object can be moved step by step. As piezoelectric elements are usually used for the clamping mechanism, the structure of the inchworms is rather complicated and expensive. In order to overcome the disadvantage, we invented a new mechanism named impact drive mechanism or IDM [1]. The principle of the movement of IDM is quite different from that of inchworm.
When we want to adjust the position of an object held with friction, we often knock the object with a hammer. The idea of IDM comes from this practice. A piezoelectric element can generate a controlled impulsive force to the object. An impact drive mechanism consists of three parts: a main body, a piezoelectric element and a weight as shown in Fig.1. The structure is much simpler than inchworm. The object to be moved is held along the base plane by friction. A piezoelectric element and a weight are connected to the main

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body.

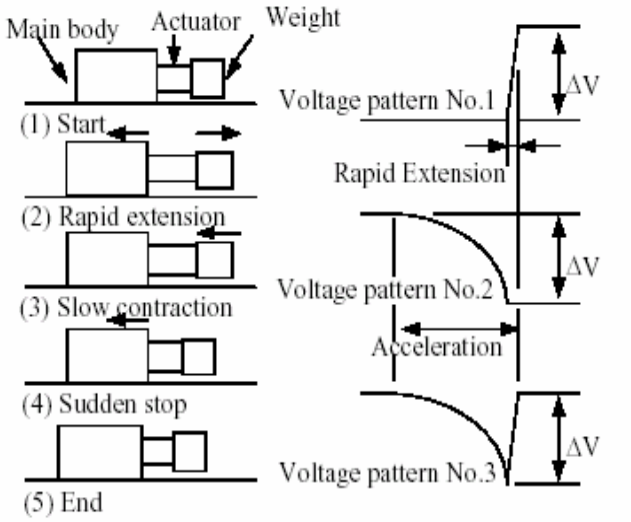


Fig. 1. Operation of impact drive mechanism.

Leftward motion is carries out in the following way as shown in Fig.1.
(1)At the beginning, the piezo is contracted.
(2)Then, the piezo is extended rapidly by the voltage pattern No.1.
An impulsive force is generated enough to push the object against the friction.
(3) After the rapid extension, the piezo is contracted with a constant acceleration by using voltage pattern No.2 so that the reaction force should not exceed the static friction to keep the displacement obtained in (2).
(4)At the end of the contraction, the deformation of the piezo is suddenly stopped. This causes another impact to the object to move leftward, and the object halts (5).
Rightward movement can be obtained only by exchanging the contraction and the extension of the piezo in each stage. In practical applications, we use the voltage pattern No.3 in Fig.1, which provides us one combined step movement of pattern No.1 and No.2 in one cycle.
By repeating the cycles, the object can move step by step. The size of the step can be ranged from several nano meters to several micrometers. Velocity of several mm/s can be obtained at several kHz of the cycles with giving a sufficient frictional force to resist the backward force.
Since the structure of IDM is very simple, it has been utilized for various positioning devices. We developed a positioning stage designed to operate in an ultra high vacuum of 10^{-12} Torr. The advantage of IDM is that the piezoelectric element can be enclosed inside the stage to shut out the outgassing from piezo to high vacuum. The hexagonal table with three pads has six piezoelectric elements with weight attached to each end. By selecting at least three piezoelectric elements, xyθ movements can be achieved in the ultra high vacuum.

Micro manipulators using IDM for insertion of a fine capillary into the elastic membrane of a cell have obtained high reputations. In addition to these applications, IDM has been applied to an electrode feeding device for a pocket size electric discharge machine, positioning stages of STM and AFM to obtain several ten nano meter step movements, and adjustment of the position of a printed board.
Following the IDM, a new kind of micro actuator named by SIDM (Smooth Impact Drive Mechanism) has been developed by Konica Minolta Co.. The principle and the picture of a SIDM are shown in Fig.2. It has been already equipped in mobile phones and cameras for auto focus mechanism and compensation of hand movement to avoid blurring of images.

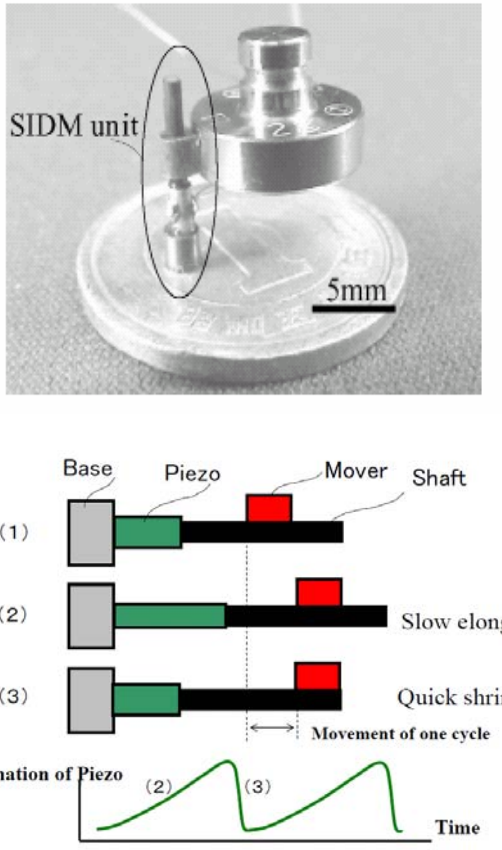


Fig. 2. SIDM and its principle.

2.2 Surface Acoustic Wave Linear Motor[2]

In ultrasonic linear motors, the slider is driven by frictional force generated between the slider and surface of traveling wave in a solid body like in Fig.3. As conventional ultrasonic motors utilizes vibration of a structure of elastic body vibrated by piezoelectric elements in resonance, the frequency of the vibration is from 10 kHz to 100 kHz, and the amplitude of the surface vibration is several μm.
Among the elastic waves passing through a solid body, the

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Rayleigh wave can exit only very thin area very close to the surface. The Rayleigh wave, which is a kind of SAW (surface acoustic wave), can be easily generated on a piezoelectric wafer by IDT (interdigital transducer) as shown in Fig.4. A couple of IDTs on a piezoelectric wafer tip can act as a high frequency electromechanical filter, which is well known as a SAW filter.

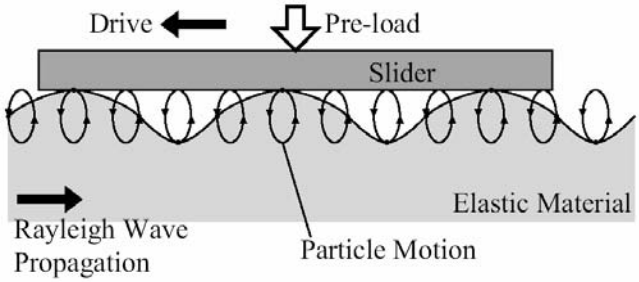


Fig. 3. Principle of SAW motor.

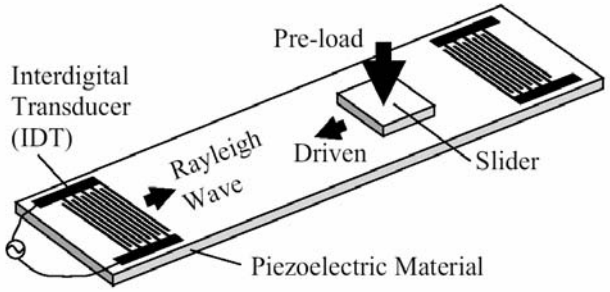


Fig. 4. Structure of SAW motor.

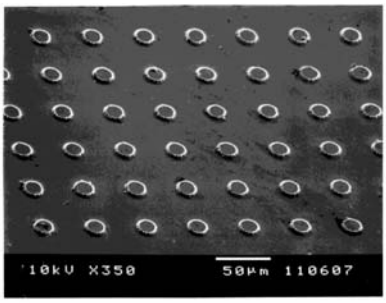


Fig.5. SEM photo of silicon slider.

Since, the particle motion of the Rayleigh wave is of the traveling wave shown in Fig.3, a new type of linear ultrasonic motor utilizing SAW can be expected. However, the characteristic of SAW is quite different from the vibration in conventional ultrasonic motors. The frequency of SAW is from 10 MHz to 100 MHz and the amplitude of the surface vibration is smaller than 20 nm. The higher frequency is favorable for precise motion control. On the other hand, there was an essential dispute about whether such a tiny amplitude as small as roughness of well finished surface could drive the slider.

First, we made a basic experimental setup and could find that small steel balls of such diameter of 0.5 mm can be driven remarkably with high acceleration in high response by supplying a sufficient contact pressure.

In order to acquire a bigger trust, a lot of contacts of high pressure should be installed on the slider. To make the multi contact point slider with much more points and less each contact area, the silicon slider with huge number of small pads fabricated by photo lithography as shown in Fig.5. The experimental results with silicon slider assures us such promising properties of the linear SAW motor as; velocity > 1 m/s, acceleration> 1000G, , response > 200kHz, positioning resolution < 10 nm, size <20mm x 5mm, thickness < 2 mm.

We expect that the SAW motor will be applied in such a precise and high response positioning system like head drive in future disc memories.

3. ELECTROSTATIC ACTUATORS

3.1 Electrostatic Powerful Flexible Motors[3]

Many kinds of micro electrostatic actuators have been developed and studied in the field of MEMS, since electrostatic drives are easily fabricated by silicon process and power density of electrostatic drives increases as its size decreases.

Though the first electrostatic drive was developed in 18th century much earlier than electromagnetic drives, the force generated in electrostatic motors was too poor to work in practical applications. By using thin films as the structure, we could develop electrostatic motors named by DEMED (Dual Excitation Multiphase Electrostatic Drive), which is strong enough to compete with electromagnetic motors. It consists of a pair of thin FPC (flexible printed circuitry) films that contain a large number of small electrodes. In each pair, one film functions as a slider and the other as a stator. As the mass of the films is very small, DEMED can produce a large power per weight ratio over 230 W/kg and over 100G acceleration.

Electrostatic motors, whose basic design is shown in Fig. 6, consist of a stator and a slider, both of which are made of a film material, and have parallel three-phase electrodes arrayed with a uniform pitch. Fig. 7 shows how the driving force is generated. By applying three-phase sinusoidal voltage to the three poles of the slider and stator, two potential distributions traveling in opposite directions are excited. The interaction between those two distributions actuates the slider. As shown in Fig.8 a high-power motor by stacking up a number of such pairs of films. In former experimental setups, the length of one side of the electrode film was between several centimeters and approximately 20 centimeters, and the pitch between electrodes was about 0.2 mm. An example of a high power device, the linear motor with 50 stacked layers of sliders and stators could develop a thrust force in excess of 300 N.

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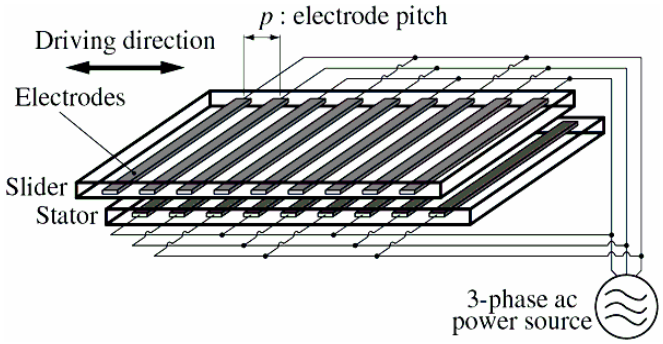


Fig. 6. Basic structure of DEMED.

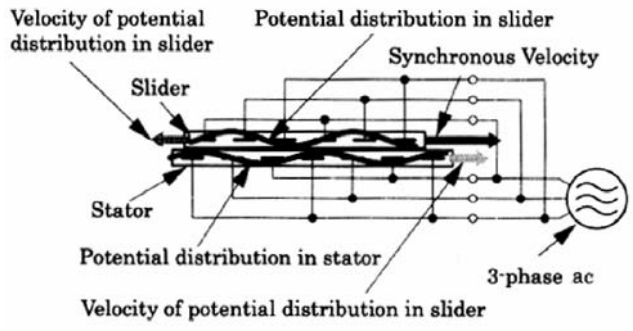


Fig. 7. Principle of drive.

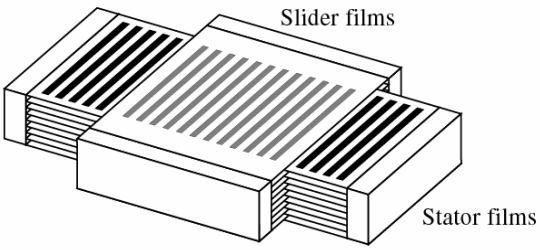


Fig.8. High power motor with stacked layers.

Since the motor consists of flexible films, it keeps the flexibility even with several layers. As shown in Fig. 9, the bent motor can operate just like the straight one.



Fig. 9. DEMED works in bending.

3.2 Particle Handling by Electric Field[4]

Electromagnetic force in linear induction motors has been applied to transportation of sheet metal and liquid metal. Such applications are to be included in the territory of linear motor. Similarly, electrostatic force can be applied to direct transportation of materials. In spite of the extremely smaller force than electromagnetic force, electrostatic force has such an important advantage that it can act upon almost all materials. We have been studying electrostatic manipulation of particles, films and thin plates, in order to provide new and useful transportation technologies in advanced factories, and clean rooms.

Traveling electric fields have been applied in mass transportation of powder by an electric curtain[9], where fine particles are levitated and against the gravity. In order to provide such devices which can convey particles more precisely and with more flexibility of manipulation like sorting, several devices have been developed. The characteristic of the drive methods is that particles are not levitated but conveyed along or very close to the surface of cover film which insulates the electrodes.

Balanced, multi-phase voltages are supplied to a series of parallel electrodes which are covered by a thin insulating film. Upon the activation, a traveling electric field is created around the electrodes. Particles on the film-surface become charged, particles move with the traveling field in a direction perpendicular to the electrodes as shown in the structure of the electric panel in Fig. 10.

By this transportation process, particles proceed stepwise from electrode to electrode, producing an overall conveyance velocity in direct proportion to the frequency and the electrode gap.

Various particle materials with a 5μm to 400μm diameter range have been examined in experiments; Fe, Al, glass, and plastic spheres showed the best performances.

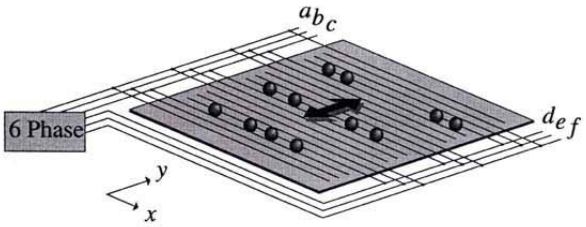


Fig. 10. Electric panel device.

An approach to achieving two-dimensional particle handling is shown in Fig.11. A mesh electric panel, which interweaves the insulated electrode wires for X drive with those for Y drive, can convey groups of particles in X and (or) Y direction.

An electric tube device with 6 phases is shown in Fig.12. The particles can be transported along the inside surface of the tube. By this device, small aluminum balls with 0.2mm diameter could be easily conveyed upward against the gravity. Therefore, this device is also suitable for long distance transportation of particles.

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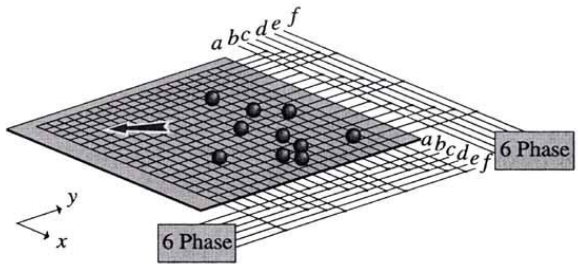


Fig. 11. Electric panel for XY manipulation.



Fig.12. An electric tube device.

Based on the electrostatic manipulation of particles, similar devices can be applied in manipulations of droplets in liquid. By controlling the voltages of each electrode, droplets can be moved independently as shown in Fig. 13. The device gives us new method for micro-TAS or Lab on a tip. By using our device, micro cannels are not necessary. In addition to the droplet manipulation, bubbles in a liquid can be manipulated just like the particles.[5]

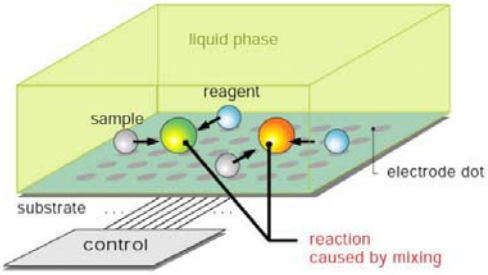


Fig. 13. Electrostatic manipulation of droplets in liquid.

3.3 Levitation and Drive of Thin Plates[6]

The valuable approach to expand the applications of liner electrostatic drive is to reduce or escape from the problem caused by the friction between the slider and stator. Therefore, by using the electrostatic levitation, where no mechanical friction exists, we can use the electrostatic force to drive the suspended object apparently.

By controlling the voltages of electrodes to keep the position and attitude of the silicon wafer, the wafer can be suspended stably without any mechanical contacts. With the same system, 300 mm wafer could be suspended with a gap of 0.4 mm.

We already achieved the electrostatic levitation of 400 mm wafer by using decentralized control for the suspension of a flexible object.

Since silicon wafers are able to be dealt as of conductive material in electrostatic suspension and drive, they can be propelled not by the induction drive but by the variable capacitance drive.

Clean transportation system for a thinner and wider plate of glass is demanded in the production of flat panel displays. Since glass has high resistivity, we did not expect to levitate a glass plate by electrostatic force without spattering of a thin conductive layer. Nevertheless, we could succeed to suspend a flat glass of 0.7mm thick by developing the electrode pattern as shown in Fig. 14. By means of the multi-divided electrodes, a glass plate can be lift up immediately after the start of control. As for the drive, glass is very suitable for the electrostatic induction drive. We already succeeded to suspend and move a 100 mm x 100 mm glass plate.

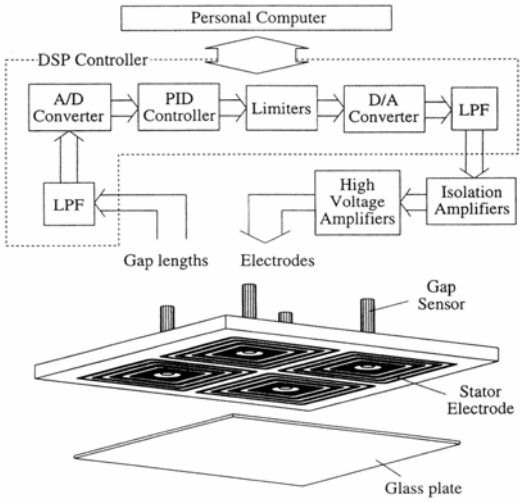


Fig. 14. Electrostatic suspension system for glass.

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